Supplement — Construction of the *Methanococcus maripaludis*, *Escherichia coli* and *Synechocystis* sp. PCC 6803 etc-GEMs, the Sharpe–Schoolfield E_a audit, and robustness

Gabriel Yvon-Durocher

Living Systems Institute, University of Exeter, Stocker Road, Exeter EX4 4QD, UK

2026-07-13

Table of contents

This Supplement documents the full construction of **all three** etc-GEMs — the *M. maripaludis* methanogenesis model, the *E. coli* respiration model and the *Synechocystis* 6803 photosynthesis model, each the first published for its taxon — together with the window-independence audit that motivates the Sharpe–Schoolfield E, the calibration diagnostics, and the robustness analyses. Numbers are model outputs; honest caveats are stated throughout.

1 Building the *E. coli* respiration etc-GEM

Base network and enzyme constraint. The base is iML1515, the manually curated *E. coli* K-12 MG1655 genome-scale reconstruction (2,712 reactions, 1,877 metabolites, 1,516 genes)[1], converted to the enzyme-constrained ecModel **eciML1515** with the GECKO method[2,3], which adds a protein-usage pseudo-metabolite per reaction drawn from a shared pool. The metabolic pool is set from independent data ($P_{\text{metab}} = \sigma\, f_{\text{metab}}\, P^{\text{lit}}$; literature protein content $P^{\text{lit}} = 0.5$ g gDW ^{1}, measured metabolic fraction, in-vivo saturation $\sigma = 0.45$[4,5]), so the growth-rate magnitude is emergent, not fit.

Model curation — uncosted O sinks. Four reactions in eciML1515 consume O at negligible enzyme cost — two quinol/menaquinol monooxygenases (QMO2, QMO3), a malate:O oxidoreductase (MOX) and a cuprous oxidase (CU1Opp) — forming a futile O cycle that at low temperature can carry the majority of gross O turnover and short-circuits the enzyme-costed cytochrome oxidase. Closing these four (a documented, reversible default) routes O through the genuine respiratory chain at ~–0.3% growth, so control is attributed to real respiration. Catalase and superoxide dismutase are real enzymes and are deliberately *not* closed.

Thermal + allocation layers. Per-enzyme T_{opt} and T_m were grounded from a sequence-based optimum predictor[6] and meltome/length relations for ~90% of enzymes (the rest at dataset means), with DLTKcat[7] refining T_{opt} where it fits; $k_{\text{cat}}(T)$ follows MMRT[8] with a literature ΔC_p prior, and the high-T collapse follows a two-state unfolding native fraction keyed on T_m. The proteome was partitioned into metabolic/biosynthesis/maintenance sectors[9,10] from a growth-condition-matched measured *E. coli* proteome, with the coupled bacterial growth law (Scott slope) under which the ribosomal fraction rises with growth rate. Maintenance is a temperature-dependent NGAM(T).

Validation and calibration. The emergent model was validated against, then Bayesian-calibrated (emcee, exact Gaussian discrepancy likelihood) to, the rich-medium (BHI) growth TPC of[11] (7–46 °C, $r_{\text{max}} = 2.40$ h ^{1} at 40 °C). The calibration frees only a few global scalars (the grounded per-enzyme parameters are not re-fit) and closes ~89% of the magnitude gap; the posterior in-vivo saturation railed toward its physical ceiling ($\sigma = 0.867$ [0.671, 0.986], above the 0.4–0.5 literature range — a genuine residual, kept distinct rather than papered over), with a melting-temperature shift ($dT_m = -5.6$ K) pulling the hot collapse to the data (Figure 1). The calibrated E_a holds near the metabolic benchmark. This is the calibrated model dissected in the main text.

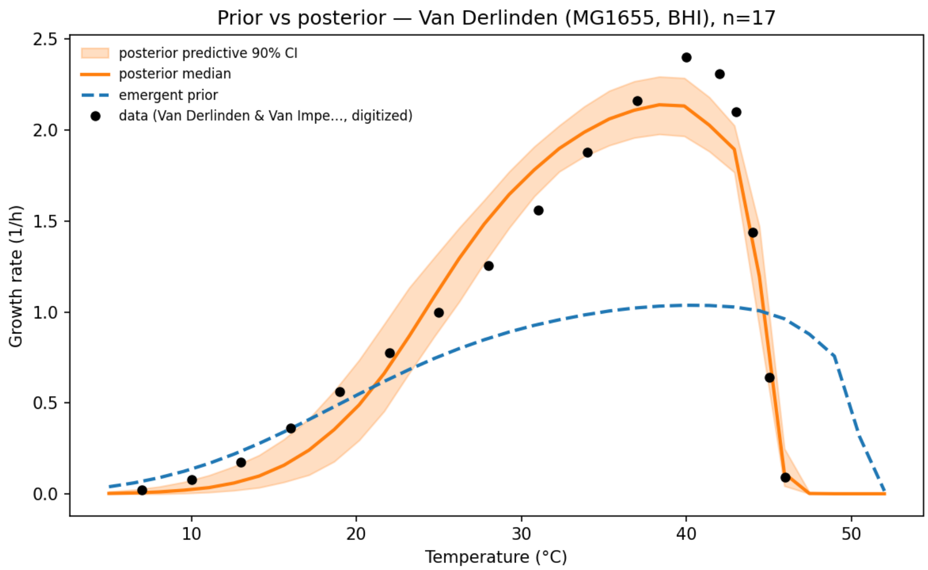

Fig. 1 | Prior (emergent) vs posterior *E. coli* TPC against the Van Derlinden 2012 BHI data.

2 Building the *M. maripaludis* methanogenesis etc-GEM (M1–M6)

The model was built in six phases, each committed and validated (build records in `strains/mmaripaludis/outputs/M*.md`).

2.1 M1 — base GEM + medium (foundation)

The iMR539 reconstruction[12] (688 reactions / 710 metabolites / 539 genes; BioModels BIOMD0000001099) was acquired and validated on a defined H2/CO2 medium. Plain FBA reproduces hydrogenotrophic methanogenesis with the correct **H2:CO2:CH4 = 4:1:1** stoichiometry and a gene-annotated Wolfe cycle. A preliminary audit found **no thermodynamic energy loops**; 85% metabolic GPR coverage; MMP locus tags map to UniProt proteome UP000000590 (sequences fetchable) — ecModel-ready.

2.2 M1b — carbon-honest curation

The published reconstruction could not grow without free uptake of four biomass precursors. Tracing the network, three (Membrane_lipid, NAC, Flagellin) feed *only* the archaellin (flagellum) glycoprotein — a dispensable motility structure — so archaellin was folded out of biomass; the fourth (tRNA(SeCys)) was a **recycling gap** (the incorporation step failed to release the free tRNA), fixed by regenerating it. Organic-carbon leaks were closed and the pinned

CH_4 exchange freed. The curated model grows autotrophically on H_2/CO_2 with **99.98% of biomass carbon from CO_2**. Two vitamin cofactors (thiamin, NMN) lack biosynthesis and are supplied as documented medium traces (0.02% of carbon).

2.3 M2 / M2b — the sMOMENT ecModel + kcat refinement

An sMOMENT total-protein pool was attached (reusing the E. coli `enzyme_cost` engine): each reaction costs MW/k_{cat} from a shared budget. Per-reaction k_{cat} by source priority: **measured core** (electron-bifurcation complexes from Milton et al. 2018, 25–165 U/mg[13]; classic biochemistry for Mtr/Fwd/Mcr/etc.), **DLTKcat** sequence predictions for the bulk, and a mean fallback. Mcr — the famously slow terminal enzyme and the E_a target — was sourced carefully (mature-Mcr average specific activity ~20 U/mg, *Methanothermobacter*; k_{cat} 59 s^{1}, uncertainty range 3–294 s^{1}). **M2b** corrected a systematic DLTKcat archaeal-underprediction (28% of predictions <1 s^{1}) with cited literature/BRENDA overrides for the flagged high-pool offenders (pyruvate- and 2-oxoglutarate:ferredoxin oxidoreductases) plus a documented 1 s^{1} floor. The grounded pool ($P_{\mathrm{total}} \times \sigma \times f_{\mathrm{metab}}$, not tuned to μ_{max}) gives an emergent μ that undershoots the observed rate ~4×, expected for a-priori kinetics.

2.4 M3 — thermal envelope + measured maintenance

kcat(T) (MMRT/Eyring) × native fraction $f_N(T)$ (two-state unfolding keyed on T_m), with per-enzyme optima from a documented **mesophile Topt prior** and T_m from a mesophile prior (the sequence/meltome predictors are the future upgrade). Maintenance is a temperature-dependent NGAM(T) **anchored on the measured** M. maripaludis maintenance energy (7.836 mmol ATP gDW^{1} h^{1}, Goyal et al. 2015[14]). The emergent TPC matches Jones' **CTmax (~47 °C)** a-priori (Figure 2); the rising limb was too warm/steep and the magnitude maintenance-crushed — the residuals the calibration then addresses.

2.5 M4 — Bayesian calibration to Jones 1983

The thermal ecModel was calibrated to the digitised Jones 1983 growth TPC[15] by ensemble MCMC (emcee; the same machinery as the E. coli fit), magnitude-first, with provenance priors. The posterior **reproduces the whole curve** (Figure 3; RMSE 0.005 h^{1}), reaching the peak (0.184 vs 0.181 h^{1}) with Topt 37 °C and CTmax 51 °C. The demanded magnitude correction (kcat_scale 7×) is a plausible in-vitro→in-vivo + archaeal-underprediction gap (the single sMOMENT pool lumps the degenerate magnitude levers). The maintenance multiplier stayed near 1, confirming the Goyal-measured NGAM.

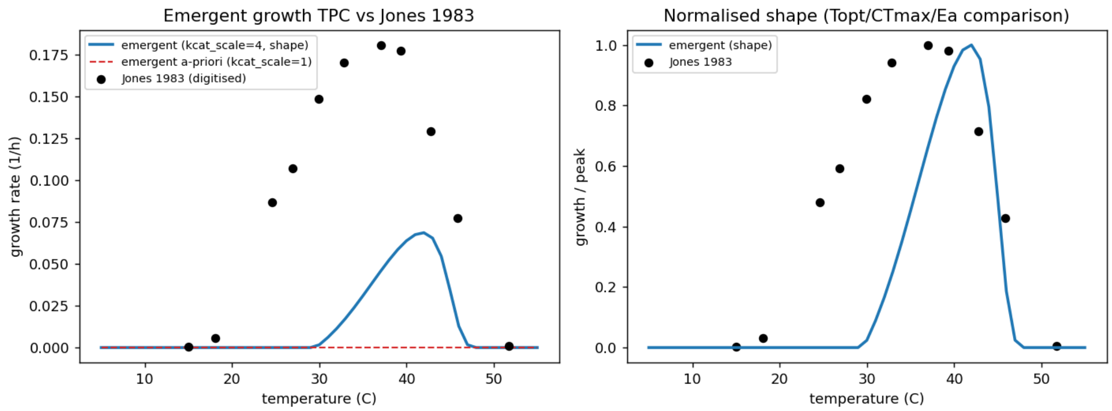

Fig. 2 | Emergent (nothing-fit) methanogen TPC vs Jones 1983: the falling limb (CTmax) matches a-priori; the rising limb and magnitude are calibrated in M4.

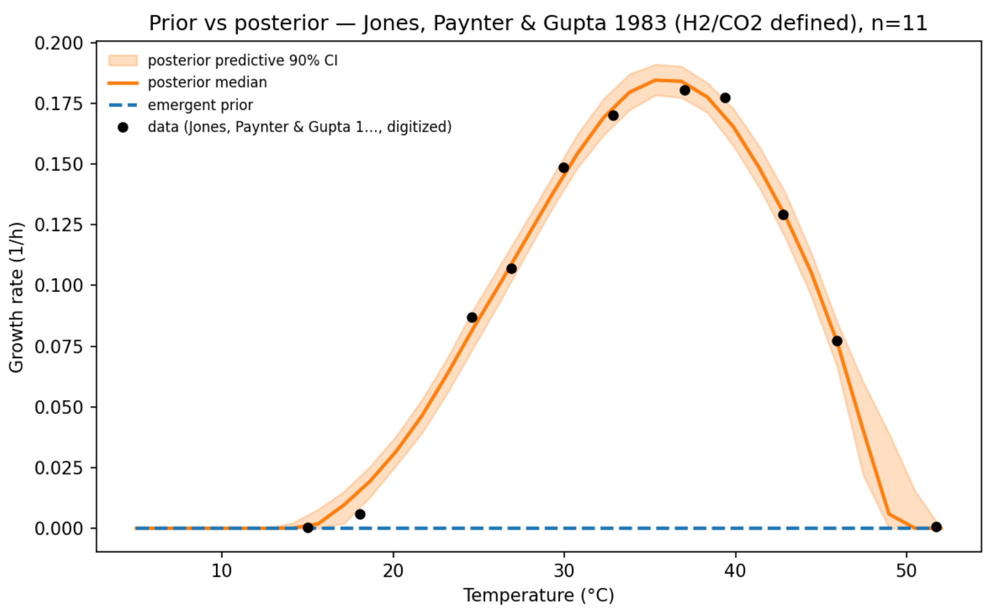

Fig. 3 | Prior (emergent) vs posterior methanogen TPC against Jones 1983.

2.6 M6 — the methanogen's own allocation layer

A proteome-sector/growth-law layer was added, grounded in the organism's **own** allocation physiology: Müller et al. 2021[16] show M. maripaludis holds a **constant ribosomal proteome fraction** and invariant catabolic/anabolic allocation across growth rate (the opposite of the E. coli/yeast scaling law), so the growth-law coupling slope is ~**0 (flat)**. Sector fractions (f_{metab} 0.55 / f_{maint} 0.30 / f_{bio} 0.15) are grounded in the Xia/Müller proteome. Forward-check: with the sector layer on and the M4 posterior, the TPC is **identical** to the single-pool model ($\max|\Delta\mu| = 0$), so no re-calibration was needed. The directly-measured allocation buffer is **+0.000 eV** (vs E. coli −0.130), grounding the allocation asymmetry in each organism's measured strategy.

Honest caveats carried into the analysis. (i) The **Mcr** $k_{\mathbf{cat}}$ (3–294 s^{1}) is the dominant single-parameter uncertainty. (ii) **DLTKcat is bacteria-trained**, so archaeal predictions are corrected but remain a prior for the biosynthetic bulk. (iii) The T_m **and Topt** are mesophile priors (no archaeal sequence predictor). (iv) The **sector fractions** are grounded but approximate; the flat slope is the robust, cited quantity. The main-text conclusions are shown robust to the parameter that matters most (Mcr; Section 6).

3 Building the *Synechocystis* 6803 photosynthesis etc-GEM (P1–P4)

The phototroph was built in four phases (build records in `strains/syn6803/outputs/P*.md`), in the **light-saturated, carbon-fixation-limited** regime chosen so that the temperature-insensitive light reactions are non-limiting and Calvin-cycle enzyme kinetics set the rate — keeping the phototroph *within* the shared enzyme-kinetic mechanism rather than requiring a new light-supply layer.

3.1 P1 — scaffold + light-saturated autotrophic base

The base reconstruction iSynCJ816[17] (1,044 reactions / 928 metabolites / 816 genes; BiGG) was acquired and a **pre-built enzyme-constrained ecModel**, iSynCJ816_STAR[18] — an AUTOPACMEN sMOMENT total-protein-pool layer (`prot_pool`, budget 0.26 g gDW^{1}, 1,066 flux-costed reactions, BRENDA/SABIO-RK kcats) — was adopted as the enzyme layer (the cyanobacterial analog of the pre-built *E. coli* ecModel, so no enzyme layer is built by hand). The medium was set photoautotrophic and **light-saturated**: photon uptake non-limiting, CO /bicarbonate available, organic carbon closed, objective the autotrophic biomass reaction. A decisive check confirmed the in-mechanism regime: in the enzyme-constrained model, growth **saturates** against photon supply and the **photon shadow price is zero** (carbon-fixation capacity, not light, is limiting), with RuBisCO carrying flux and zero net photorespiration — impossible in a plain (non-enzyme) model, which is photon-linear.

3.2 P2 — thermal envelope + measured maintenance

The temperature-dependent envelope — kcat(T) (MMRT/Eyring) × native fraction $f_N(T)$ (two-state unfolding keyed on T_m) — was attached to iSynCJ816_STAR by the same machinery as the other two models. Per-enzyme optima and melting temperatures use the documented **mesophile prior** (6803 is a mesophile absent from the sequence/meltome training sets; the same route as the methanogen), the curvature the literature MMRT prior. Maintenance NGAM(T) is anchored on the **measured** *Synechocystis* photon-maintenance coefficient (3.12 mmol gDW ^{1} h ^{1}; Touloupakis et al. 2015[19]) — a photon-based coefficient, reflecting that maintenance ATP is met by photophosphorylation. The emergent (nothing-fit) TPC already matches shape and magnitude a-priori (Topt 36 °C vs observed ~35; CTmax 46 °C; $r_{\max}$ 0.067 h ^{1}, realistic for 6803; Figure 4), with one clear residual — the rising-limb E ran to 0.77 eV versus the observed ~0.5 — which the calibration then addresses.

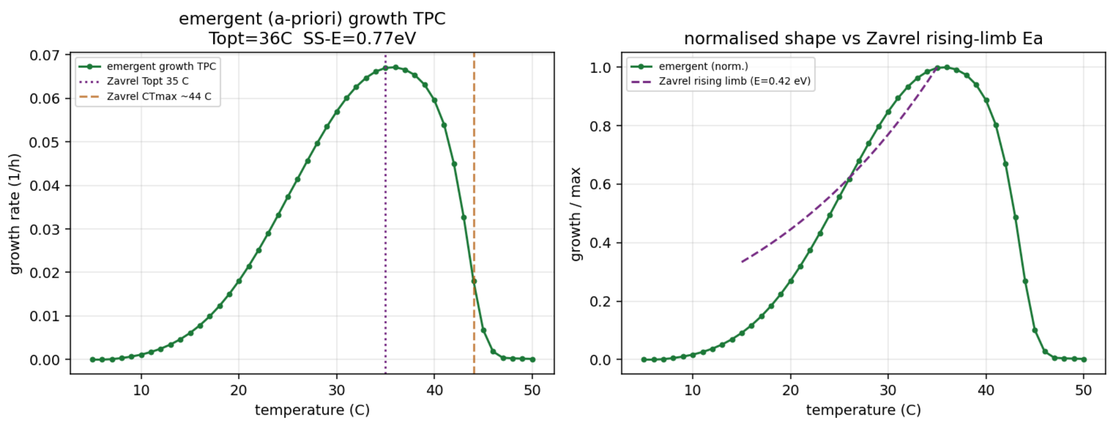

Fig. 4 | Emergent (nothing-fit) *Synechocystis* TPC vs Zavrel 2015 (light-saturated growth): shape and magnitude match a-priori; the rising-limb Ea is calibrated in P3.

3.3 P3 — Bayesian calibration to Zavrel 2015 (shape-first)

The thermal ecModel was calibrated to the digitised[20] light-saturated growth TPC by ensemble MCMC (emcee; the same machinery as the other two), **shape-first**: the a-priori magnitude is already realistic, so `kcat_scale` was centred on 1 (not the methanogen's Davidi-4× prior) and the envelope knobs did the work. The posterior reproduces the curve (Figure 5; $r_{\max}$ 0.094 vs 0.097 h ^{1}, Topt 35 °C) with **modest, plausible** corrections — `kcat_scale` ×1.22 and a curvature `dCp_scale` ×0.85 (i.e. −3.4 vs the −4 kJ mol ^{1} K ^{1} prior; **not** railed to an implausible floor). The **Ea-lever diagnostic** is the key P3 result: the low photosynthesis E_a is achieved with the curvature knob at essentially its prior and **no allocation buffer needed**, so it is genuinely **shallow Calvin-cycle kinetics**, not an artefact. The observed Zavrel growth-E (0.44) is fragile — a ±5% rate perturbation spans a 90% range [0.34, 1.01], because six points over a narrow window under-determine the metric — so the robust anchor is the window-independent model E (0.57) and the independent Inoue photosynthesis-flux E (0.52), which the model's carbon-fixation flux E (0.56) matches (Figure 6). The Inoue *light-limited* growth curve (Topt ~25 °C) is shown as the light-limitation-scope cross-check, not a fit target.

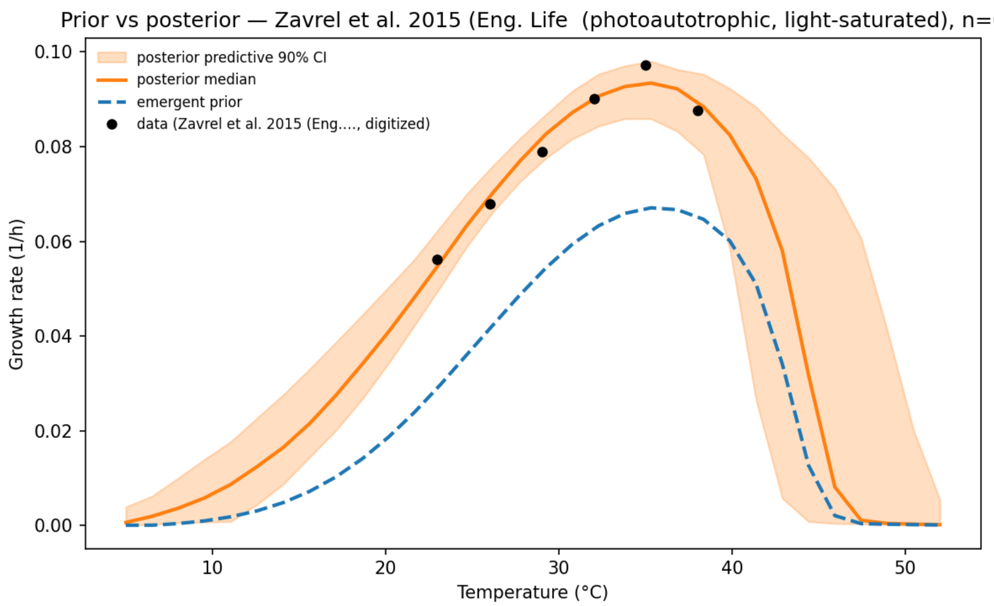

Fig. 5 | Prior (emergent) vs posterior *Synechocystis* TPC against Zavrel 2015.

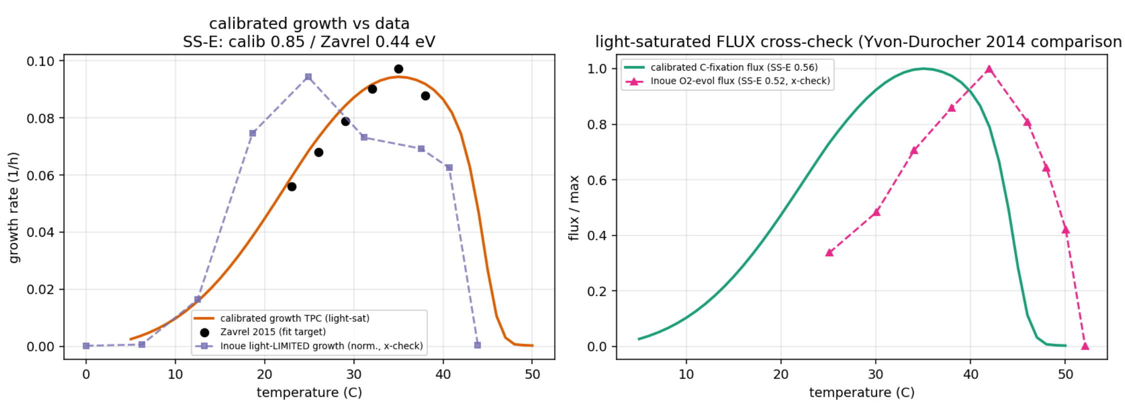

Fig. 6 | Independent cross-checks: the calibrated carbon-fixation-flux Ea matches the Inoue 2001 O -evolution flux Ea (0.56 vs 0.52; right), and the Inoue light-limited growth curve peaks lower (Topt ~25 °C), illustrating the light-limitation scope boundary (left).

3.4 P3b — the phototroph's own allocation layer

A proteome-sector/growth-law layer was added, grounded in the cyanobacterium's **own** allocation physiology: Jahn et al. 2018[21] resolve seven proteome sectors versus growth rate (total protein invariant), which map onto our three as f_{metab} 0.50 (light-harvesting + photosystems + carbon fixation + metabolism), f_{maint} 0.33 (MAI), f_{bio} 0.17 (RIB, rising with growth rate = the coupling slope), corroborated by Zavrel et al. 2019[22] and the Faizi et al. 2018 optimal-allocation framework[23], with the heat-shock response[24] informing the hot-side maintenance sector. Because *Synechocystis*'s ribosome fraction is nearly constant and its growth-rate range is ~10–20× narrower than *E. coli*'s, the ribosome cap is pinned non-binding and the small allocation buffer comes from the metabolic-sector shrinkage. Forward-check: with the layer on and the P3 posterior, the TPC is near-identical (Topt/CTmax preserved, $r_{\max}$ −4%, max$|\Delta\mu|$ 0.004; Figure 7), so no re-calibration was needed. The directly-measured allocation buffer is **−0.019 eV** — small, between *E. coli*'s −0.13 (Scott) and the methanogen's 0.00 (Müller-flat), confirming on a measured footing that the low photosynthesis E_a is kinetic, not allocation-set.

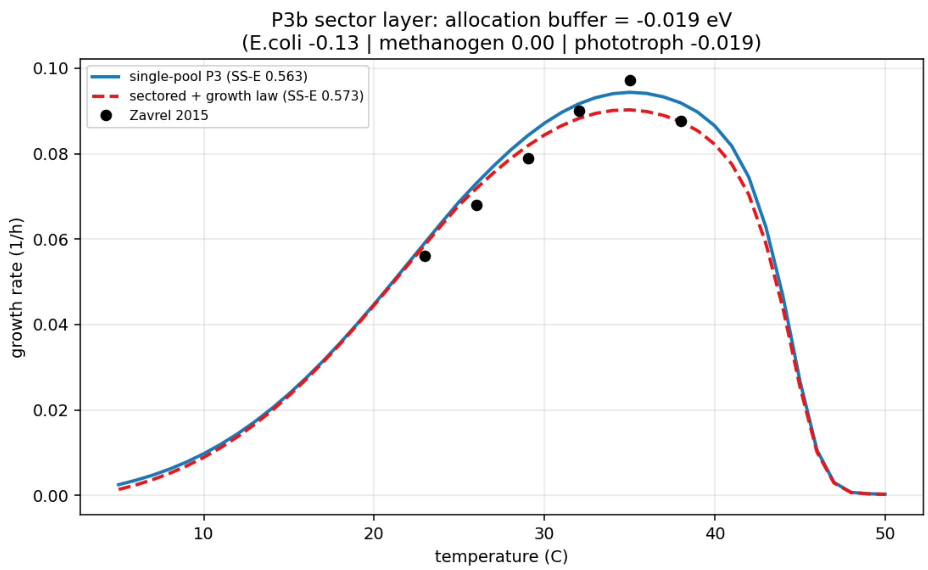

Fig. 7 | Sectored vs single-pool *Synechocystis* TPC: the Jahn shallow-ribosome allocation layer is near-neutral (allocation buffer −0.02 eV).

3.5 P4 — dissection + robustness

The calibrated, sectored SS-E (0.573 eV) was decomposed identically to the other two organisms (main text **?@sec-decomp**): naive 0.684 + control −0.016 + allocation −0.019 + maintenance −0.027 + aggregation −0.050. The carbon-fixation/Calvin enzymes (transketolase, aldolase, RuBisCO, phosphoribulokinase, fructose-bisphosphatase) carry 34% of the control term at mean E 0.63, below the 0.68 proteome mean — a **low-E_a controlling backbone**, the mirror image of the methanogen. Robustness (Figure 8): sweeping RuBisCO k_{cat} (0.5–4×) and the whole Calvin backbone (0.5–2×) leaves the SS-E within [0.565, 0.577] — the low E_a is a curvature/shape property, not a kcat-level artefact — while the shared MMRT curvature prior (−2 to −6 kJ mol^{-1} K^{-1}) scales all three organisms together.

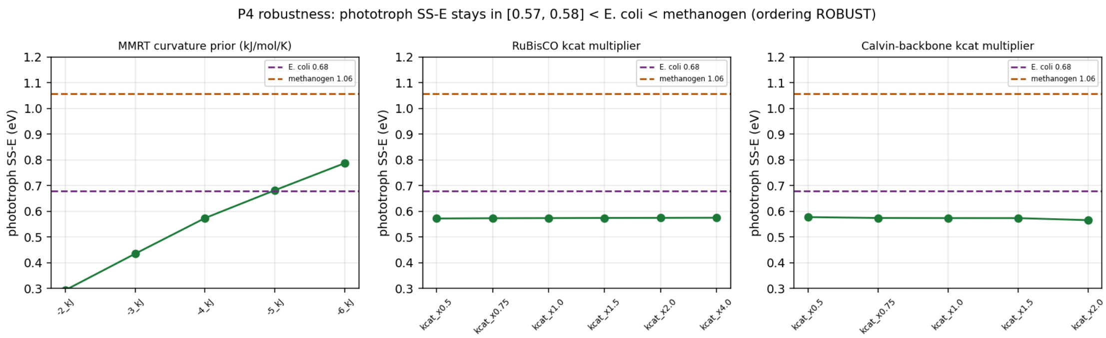

Fig. 8 | Robustness of the phototroph SS-E to the carbon-fixation kcat (invariant) and the shared dCp prior; the ordering below E. coli and the methanogen is robust to the phototroph's own kinetic uncertainty.

Honest caveats carried into the analysis. (i) The observed **Zavrel growth-E is fragile** (6 points, narrow window); the robust anchor is the model E (0.57) and the independent Inoue flux E (0.52). (ii) **Topt and T_m** are mesophile priors (no cyanobacterial sequence/meltome predictor run). (iii) The result is for the **light-saturated** regime; light limitation lowers the E_a further through a non-enzymatic (photon-supply) route outside the present mechanism (Discussion). (iv) **Photorespiration** temperature-dependence is not modelled (suppressed under the CO_2-replete calibration conditions). (v) The **sector fractions** are grounded but approximate; the small buffer is the robust, cited quantity.

4 Parameter provenance and the global correction knobs

Every per-enzyme and whole-cell input is grounded in independent data or a stated literature value; nothing is fit to the growth curve in the emergent model. Table 1, Table 2 and Table 3 give the provenance for each organism, and Table 4 the global scalars freed in calibration (each transforms the whole grounded per-enzyme distribution while preserving its shape; all are no-ops at their nominal).

Table 1: *E. coli* parameter provenance — grounded in independent data, never fit to growth.

Symbol	Quantity	Source	Coverage / value
$\mathrm{MW}_i, k_{\mathrm{cat,ref},i}$	enzyme mass, reference turnover	GECKO ecModel of iML1515[1–3]	2,560 enzymes
$T_{\mathrm{opt},i}$	catalytic optimum	sequence predictor[6] + DLTKcat[7]	~90% grounded
$T_{m,i}$	melting temperature	measured *E. coli* meltome[25]	~90% grounded; rest 55.6 °C
$\Delta C^{\ddagger}_{p,i}$	turnover curvature	literature MMRT prior[8]	−4 kJ mol ^{1} K ^{1}
$f_{\mathrm{metab}}/f_{\mathrm{bio}}/f_{\mathrm{maint}}$	sector fractions	measured proteome per medium & T[26]	0.48/0.19/0.33 (glucose 30 °C)
s	growth-law slope	Scott 2010[10]	0.33 (rising ribosome fraction)
P^{lit}, σ	total protein, saturation	literature[4,5]	0.5 g gDW ^{1}, 0.45
NGAM(T)	maintenance	*E. coli* maintenance port[27]	a_{m} 8.5, E_m 0.5 eV
validation	growth TPC	K-12 MG1655, BHI, 7–46 °C[11]	1 curve

Table 2: *M. maripaludis* parameter provenance. The measured energy-core kinetics, the measured maintenance, and the constant-ribosome allocation law are the organism-specific grounded inputs.

Symbol	Quantity	Source	Coverage / value
MW_i	enzyme mass	UniProt proteome UP000000590 (from GPRs)	527/527 enzymatic reactions
$k_{\mathrm{cat},i}$	turnover	measured core[13] + DLTKcat[7] + literature overrides + 1 s ^{1} floor	11 core / 293 DLTKcat / 223 fallback
k_{cat}(Mcr)	terminal enzyme	mature-Mcr specific activity	58.9 s ^{1} (range 3–294; swept)
$T_{\mathrm{opt},i}, T_{m,i}$	optimum, stability	mesophile prior (sequence/meltome predictors not run)	T_{opt} ~39 °C, T_m ~55.6 °C
$\Delta C^{\ddagger}_{p,i}$	turnover curvature	literature MMRT prior[8]	−4 kJ mol ^{1} K ^{1}

Symbol	Quantity	Source	Coverage / value
$f_{\text{metab}}/f_{\text{bio}}/f_{\text{maint}}$	sector fractions	Xia/Müller proteome[16]	0.55/0.15/0.30
s	growth-law slope	Müller 2021 constant ribosome fraction[16]	0 (flat law)
P^{lit}, σ	total protein, saturation	literature[4,5]	0.5 g gDW ^{1}, 0.45
NGAM(T)	maintenance	**measured** *M. maripaludis*[14]	7.836 mmol ATP gDW ^{1} h ^{1} @ 37 °C
validation	growth TPC	type strain JJ, H /CO [15]	1 curve

Table 3: *Synechocystis* 6803 parameter provenance. The pre-built ecModel kinetics (with well-measured RuBisCO/Calvin turnovers), the measured photon-maintenance, and the Jahn shallow-ribosome allocation law are the organism-specific grounded inputs.

Symbol	Quantity	Source	Coverage / value
$\text{MW}_i, k_{\text{cat,ref},i}$	enzyme mass, reference turnover	pre-built ecModel iSynCJ816_STAR (AUTOPACMEN sMOMENT)[17,18]	1,066 flux-costed reactions; BRENDA/SABIO-RK
k_{cat}(RuBisCO/Calvin)	carbon-fixation core	ecModel (well-measured); swept for robustness	RuBisCO 0.5–4×, Calvin 0.5–2×
$T_{\text{opt},i}, T_{m,i}$	optimum, stability	mesophile prior (sequence/meltome predictors not run)	T_{opt} ~39 °C, T_m ~55.6 °C
$\Delta C^{\ddagger}_{p,i}$	turnover curvature	literature MMRT prior[8]	−4 kJ mol ^{1} K ^{1}
$f_{\text{metab}}/f_{\text{bio}}/f_{\text{maint}}$	sector fractions	Jahn 2018[21] + Zavrel 2019[22] + Faizi 2018[23]	0.50/0.17/0.33
s	growth-law slope	Jahn 2018 RIB vs μ[21]	weak rise, small over $\mu \leq 0.11$ h ^{1}
P^{lit}, σ	total protein, saturation	Zavrel 2019[22];[4,5]	~0.4 g gDW ^{1}, 0.45
NGAM(T)	maintenance	**measured** photon-maintenance[19]; heat-shock[24]	3.12 mmol gDW ^{1} h ^{1}
validation	growth + flux TPC	light-sat. autotrophy[20]; O -evolution flux[28]	2 curves

Table 4: Global correction knobs freed in calibration — each is one scalar acting on the whole per-enzyme distribution.

Knob	Transformation	Effect
dT_{opt}	$T_{\mathrm{opt},i} \to T_{\mathrm{opt},i} + dT_{\mathrm{opt}}$	uniform shift of every optimum
`topt_scale`	$T_{\mathrm{opt},i} \to T_0 + \texttt{topt\_scale}(T_{\mathrm{opt},i} - T_0)$	spread of optima
`dCp_scale`	$\Delta C^{\ddagger}_{p,i} \to \texttt{dCp\_scale}\, \Delta C^{\ddagger}_{p,i}$	turnover curvature / breadth
dT_m	evaluate $f_{N,i}$ at $T - dT_m$	uniform melting shift (+ raises CT_{max})
`tm_scale`	$T_{m,i} \to \bar{T}_m + \texttt{tm\_scale}(T_{m,i} - \bar{T}_m)$	spread of melting temperatures
`kcat_scale` (λ_k)	$c_i \to c_i/\lambda_k$	global turnover level $\to r_{\mathrm{max}}$
$f_{\mathrm{metab}}, f_{\mathrm{maint}}$	set the sector simplex	proteome allocation
`ngam_scale`, `ngam_steepness`	scale a_{m}, E_m	maintenance amplitude / steepness
σ	scale both sector caps	in-vivo saturation (magnitude)
σ_{disc}	likelihood variance	model-discrepancy noise (nuisance)

5 Why Sharpe–Schoolfield E: the window-sensitivity audit

The naive Boltzmann–Arrhenius slope of the rising limb depends on the fitting window because the limb is curved. Table 5 recomputes it under five window conventions for every model and observed curve: the spread is **0.42–1.41 eV per curve**, larger than the effects we attribute, and enough to reverse the cross-organism ordering. The Sharpe–Schoolfield E (the high-temperature-inactivation model, main text) is read from the whole shape, is window-independent, and — the decisive test — makes model and observed agree for both organisms (Figure 9).

```
import pandas as pd
w = pd.read_csv("assets/tables/window_sensitivity.csv")
ss = pd.read_csv("assets/tables/sharpe_schoolfield_fits.csv")[["curve","SS_E_eV","SS_r2" if
m = w.merge(ss, on="curve", how="left")
print(m.round(3).to_markdown(index=False))
```

Table 5: Full window-sensitivity table: Boltzmann-Arrhenius slope Ea (eV) under five window conventions, and the window-independent Sharpe-Schoolfield E, for every model + observed curve.

curve	rel_10_95	full_below	rel_10_80	rel_30_90	rel_10_50	spread_eV	SS_E_eV	r2
ecoli_emergent	0.609	0.64	0.802	0.489	1.054	0.565	0.458	0.896
ecoli_tuned	0.956	1.468	1.203	0.756	1.56	0.804	0.677	0.93
ecoli_observed_VdL	0.643	0.963	0.683	0.545	0.839	0.418	0.558	0.973
methan_calibrated	1.174	1.462	1.372	0.892	1.785	0.893	1.024	0.957
methan_observed_Jones	0.66	2.067	nan	0.781	nan	1.407	1.042	0.95

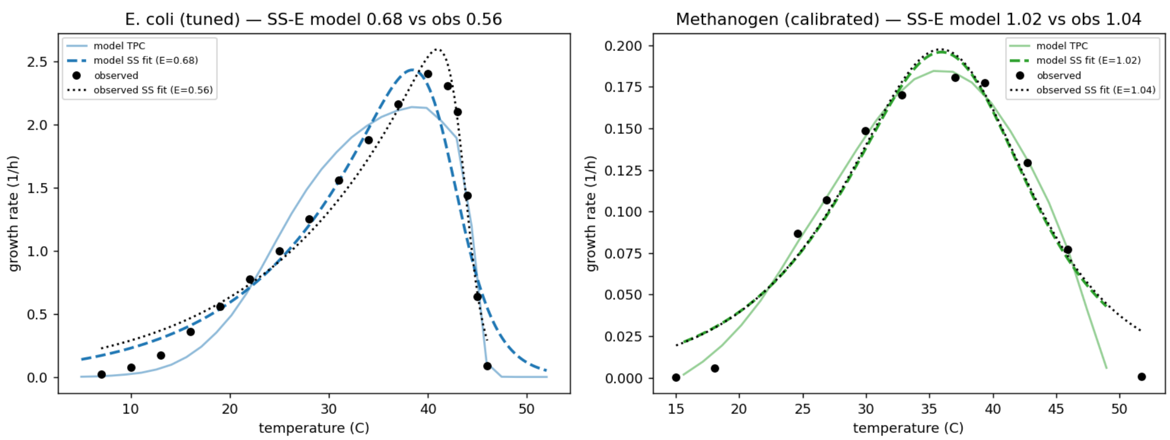

Fig. 9 | Sharpe–Schoolfield fits overlaid on the model and observed curves for both organisms; the model and observed E agree.

6 Mcr-$k_{\mathbf{cat}}$ sensitivity

Table 6 sweeps Mcr across its 3–294 s ^{1} range. The organism E_a stays **1.01–1.27 eV, always well above E. coli's 0.68**, and the methanogenesis backbone carries **83–103%** of the control term throughout — so the ordering and the backbone-control finding are robust. Only the identity of the single leading enzyme shifts (Mcr when slow → Fwd when fast).

```python
import pandas as pd
m = pd.read_csv("assets/tables/mcr_sweep.csv")
print(m.round(3).to_markdown(index=False))
```

Table 6: Mcr-kcat sensitivity sweep: organism SS-E, backbone fraction of the control term, and the leading enzymes, across 3-294/s.

Mcr_kcat	SS_E	slope	backbone_frac_of_control	top3	Mcr_C_i	Mcr_contrib
3	1.274	1.597	1.026	Mcr(1.605);Fwd(0.092);CODH(-0.065)	2.391	1.605
10	1.01	1.105	0.986	Mcr(0.876);Fwd(0.167);CODH(-0.039)	1.907	0.876
30	1.065	1.132	0.925	Mcr(0.488);Fwd(0.277);Mtr(0.059)	0.491	0.488
58.9	1.056	1.081	0.887	Fwd(0.337);Mcr(0.304);Mtr(0.072)	0.727	0.304
100	1.066	1.153	0.857	Fwd(0.395);Mcr(0.211);Mtr(0.079)	0.178	0.211
294	1.017	1.074	0.833	Fwd(0.427);Mtr(0.086);Mer(0.076)	0.766	0.078

7 Calibration diagnostics

All three models were calibrated by ensemble Markov-chain Monte Carlo (emcee) with an exact Gaussian discrepancy likelihood and autocorrelation-based early stopping; the grounded per-enzyme parameters are held fixed and only a small set of global scalars is freed. Figure 13 shows the posterior corner plots. For *E. coli* the fit demands a large in-vivo saturation ($\sigma \to$ its physical ceiling) and a modest melting-temperature shift; for *M. maripaludis* it demands a several-fold turnover magnitude correction (kcat_scale 7, absorbing the in-vitro→in-vivo gap and the residual archaeal-kcat underprediction on the single sMOMENT pool), while the maintenance multiplier stays near one — confirming the Goyal-measured NGAM; for *Synechocystis* the calibration is **shape-first** (magnitude already realistic), demanding only a small kcat_scale (×1.2) and a modest curvature reduction (dCp_scale ×0.85, not railed), the diagnostic that the low photosynthesis E_a is genuinely shallow Calvin-cycle kinetics. In all three the posterior-predictive curve passes through the observed data (the main-text Figure 1), and the resulting Sharpe–Schoolfield E_a matches the observed value.

1. Monk, J. M. *et al.* iML1515, a knowledgebase that computes Escherichia coli traits. *Nature Biotechnology* **35**, 904–908 (2017).
2. Sánchez, B. J. *et al.* Improving the phenotype predictions of a yeast genome-scale metabolic model by incorporating enzymatic constraints (GECKO). *Molecular Systems Biology* **13**, 935 (2017).

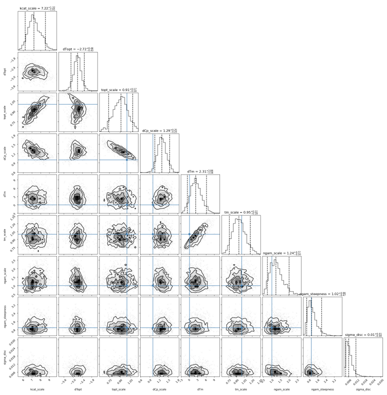

Fig. 10 | *M. maripaludis* (Jones 1983).

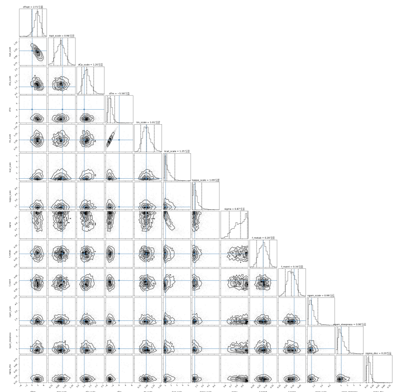

Fig. 11 | *E. coli* (Van Derlinden 2012).

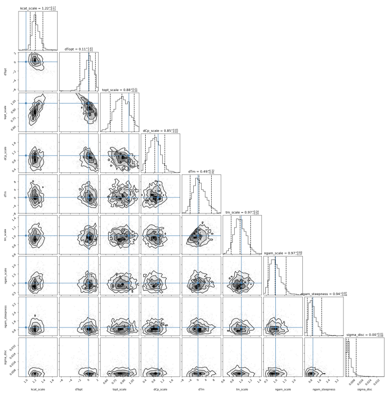

Fig. 12 | *Synechocystis* 6803 (Zavrel 2015).

Posterior corner plots for the three calibrations.

Fig. 13

239 3. Domenzain, I. *et al.* Reconstruction of a catalogue of genome-scale metabolic models with enzymatic constraints using GECKO 2.0. *Nature Communications* **13**, 3766 (2022).

240 4. Davidi, D. *et al.* Global characterization of in vivo enzyme catalytic rates and their correspondence to in vitro kcat measurements. *Proceedings of the National Academy of Sciences* **113**, 3401–3406 (2016).

241 5. Heckmann, D. *et al.* Kinetic profiling of metabolic specialists demonstrates stability and consistency of in vivo enzyme turnover numbers. *Proceedings of the National Academy of Sciences* **117**, 23182–23190 (2020).

242 6. Li, G., Rabe, K. S., Nielsen, J. & Engqvist, M. K. M. Machine learning applied to predicting microorganism growth temperatures and enzyme catalytic optima. *ACS Synthetic Biology* **8**, 1411–1420 (2019).

243 7. Qiu, S., Zhao, S. & Yang, A. DLTKcat: Deep learning-based prediction of temperature-dependent enzyme turnover rates. *Briefings in Bioinformatics* **25**, bbad506 (2024).

244 8. Hobbs, J. K. *et al.* Change in heat capacity for enzyme catalysis determines temperature dependence of enzyme catalyzed rates. *ACS Chemical Biology* **8**, 2388–2393 (2013).

245 9. Basan, M. *et al.* Overflow metabolism in Escherichia coli results from efficient proteome allocation. *Nature* **528**, 99–104 (2015).

246 10. Scott, M., Gunderson, C. W., Mateescu, E. M., Zhang, Z. & Hwa, T. Interdependence of cell growth and gene expression: Origins and consequences. *Science* **330**, 1099–1102 (2010).

247 11. Van Derlinden, E. & Van Impe, J. F. Quantifying the heterogeneous heat response of Escherichia coli under dynamic temperatures. *International Journal of Food Microbiology* **158**, 73–78 (2012).

248 12. Richards, M. A. *et al.* Exploring hydrogenotrophic methanogenesis: A genome scale metabolic reconstruction of Methanococcus maripaludis. *Journal of Bacteriology* **198**, 3379–3390 (2016).

249 13. Milton, R. D. *et al.* Measured specific activities and redox thermodynamics of the flavin-based electron-bifurcation hydrogenase and heterodisulfide-reductase complexes of Methanococcus maripaludis. *Biochemistry* (2018).

250 14. Goyal, N., Widiastuti, H., Karimi, I. A. & Zhou, Z. Flux measurements and maintenance energy for carbon dioxide utilization by Methanococcus maripaludis. *Microbial Cell Factories* **14**, 146 (2015).

251 15. Jones, W. J., Paynter, M. J. B. & Gupta, R. Characterization of Methanococcus maripaludis sp. Nov., a new methanogen isolated from salt marsh sediment. *Archives of Microbiology* **135**, 91–97 (1983).

252 16. Müller, A. L. *et al.* An alternative resource allocation strategy in the chemolithoautotrophic archaeon Methanococcus maripaludis. *Proceedings of the National Academy of Sciences* **118**, e2025854118 (2021).

253 17. Joshi, C. J., Peebles, C. A. M. & Prasad, A. Metabolic network reconstruction and analysis of synechocystis sp. PCC 6803 (iSynCJ816). *Algal Research* **27**, 295–310 (2017).

254 18. Andrews, F. iSynCJ816_STAR: An enzyme-constrained (AUTOPACMEN sMOMENT) version of the synechocystis sp. PCC 6803 model iSynCJ816. (2024).

255 19. Touloupakis, E., Cicchi, B. & Torzillo, G. A bioenergetic assessment of photosynthetic growth of synechocystis sp. PCC 6803 in continuous cultures. *Biotechnology for Biofuels* **8**, 133 (2015).

256 20. Zavřel, T., Sinetova, M. A., Červený, J., Ludwig, M. & Červený, J. Characterization of a model cyanobacterium synechocystis sp. PCC 6803 autotrophic growth in a flat-panel photobioreactor. *Engineering in Life Sciences* **15**, 122–132 (2015).

257 21. Jahn, M. *et al.* Growth of cyanobacteria is constrained by the abundance of light and carbon assimilation proteins. *Cell Reports* **25**, 478–486 (2018).

258 22. Zavřel, T. *et al.* Quantitative insights into the cyanobacterial cell economy. *eLife* **8**, e42508 (2019).

259 23. Faizi, M., Zavřel, T., Loureiro, C., Červený, J. & Steuer, R. A model of optimal protein allocation during phototrophic growth. *Biosystems* **166**, 26–36 (2018).

260 24. Suzuki, I., Simon, W. J. & Slabas, A. R. The heat shock response of synechocystis sp. PCC 6803 analysed by transcriptomics and proteomics. *Journal of Experimental Botany* **57**, 1573–1578 (2006).

261 25. Leuenberger, P. *et al.* Cell-wide analysis of protein thermal unfolding reveals determinants of thermostability. *Science* **355**, eaai7825 (2017).

262 26. Wang, D. *et al.* A proteome optimal allocation model for elucidating effects of temperature on bacterial growth. *npj Systems Biology and Applications* https://doi.org/10.1038/s41540-026-00693-4 (2026) doi:10.1038/s41540-026-00693-4.

263 27. Madkaikar, A. Predicting the temperature dependence of microbial metabolism with enzyme- and temperature-constrained genome-scale models. (Imperial College London, 2023).

28. Inoue, N. *et al.* Acclimation to the growth temperature and the high-temperature effects on photosystem II and plasma membranes in a mesophilic cyanobacterium, synechocystis sp. PCC 6803. *Plant and Cell Physiology* **42**, 1140–1148 (2001).